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10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

END SEMESTER PROJECT REVIEW

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Smart Ticket Booking Event

SDG: SDG No – SDG Title(SDG 9 – Industry, Innovation and Infrastructure)

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INTRODUCTION



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❖ In today's digital era, manual event registration and ticket booking methods lack convenience, efficiency, and real-time accessibility.

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❖ Users expect fast, secure, and user-friendly platforms to view events and book tickets online.

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❖ This project addresses the need for a structured and efficient ticket booking system.

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❖ The system allows users to log in, view event details, check ticket availability, and book tickets.

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❖ It ensures proper input validation and prevents overbooking of tickets.

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❖ The application integrates React JS, Node.js, and MySQL to manage users and bookings efficiently.

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❖ This project demonstrates real-world implementation of web technologies used in event management systems.

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❖ Overall, it provides a scalable and user-friendly solution for digital event booking platforms.

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PROBLEM STATEMENT



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❑ Many institutions still rely on manual or semi-digital methods to manage event registrations and ticket bookings.

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❑ This leads to inefficiencies in handling participant data, tracking ticket availability, and managing event records.

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❑ Manual systems increase the risk of overbooking, data errors, and delayed confirmation processes.

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❑ There is a growing need for an automated and centralized ticket booking system.

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❑ Organizers require a reliable system to manage events, users, and bookings efficiently.

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❑ Users expect secure login, easy access to event details, and a smooth ticket booking experience.

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❑ Hence, a web-based Ticket Booking System is proposed to address these challenges.

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❑ The system aims to improve accuracy, accessibility, security, and overall event management efficiency.

OBJECTIVE(S)



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- ✓ To design and develop a secure web-based Ticket Booking Management System using React, Node.js, and MySQL for efficient management of events and bookings.

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- ✓ To implement user authentication and validation to ensure secure login and accurate user input handling.

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- ✓ To provide an interactive and user-friendly interface that allows users to view event details and book tickets easily.

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- ✓ To automate ticket booking and maintain accurate records of bookings using a structured database system.

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- ✓ To reduce manual effort and improve efficiency through a centralized online event management platform.

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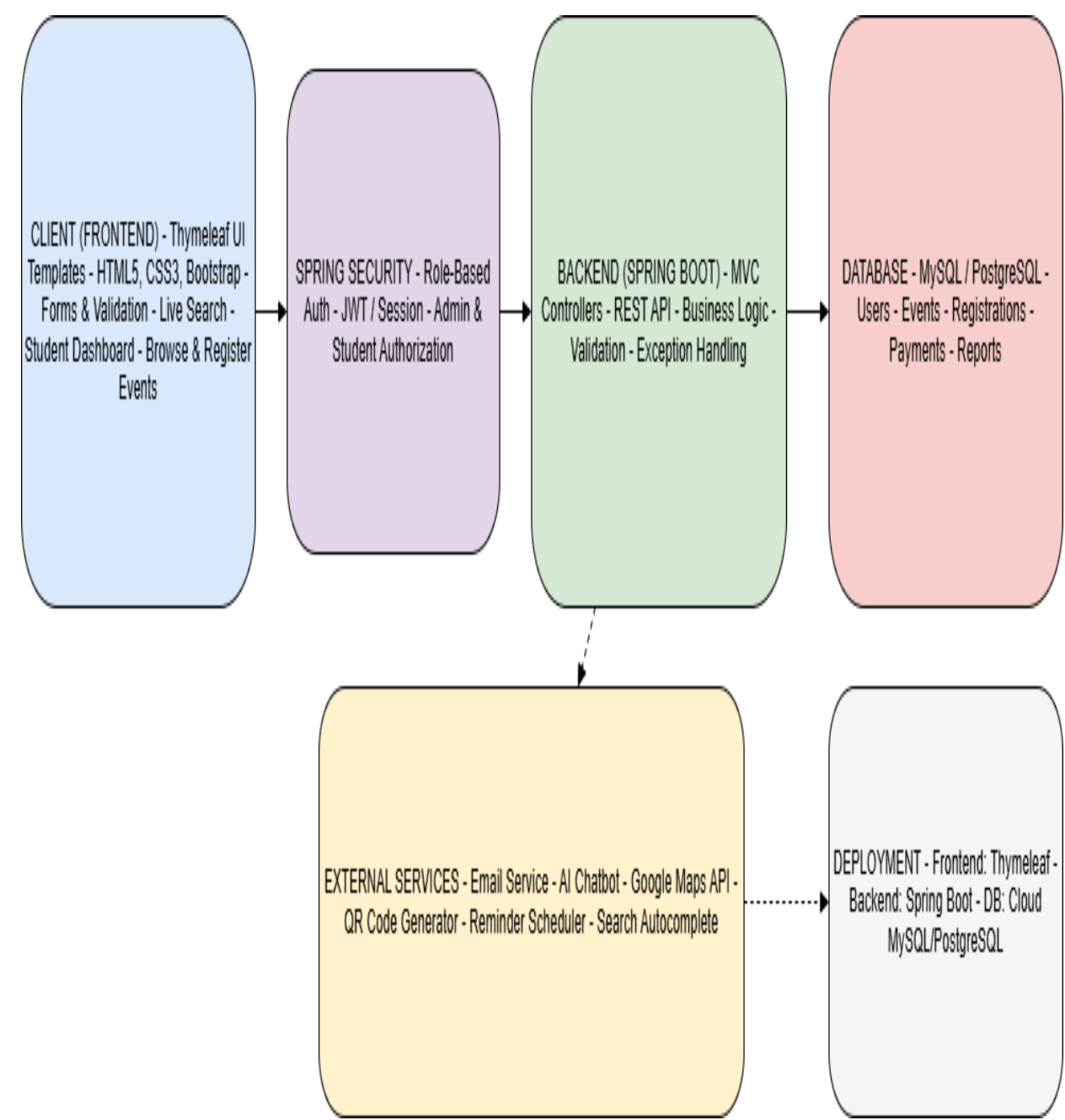
PROPOSED SYSTEM



- The proposed solution is a web-based Ticket Booking System developed using React, Node.js, and MySQL.
- The system provides secure user login and role-based access for users and admin.
- All user and booking data are stored and managed efficiently using a structured database system.
- This system solves issues of manual registration, overbooking, and data management errors found in traditional methods.
- It reduces paperwork and improves speed, accuracy, and reliability in managing event bookings.
- Compared to existing manual or semi-automated systems, it offers better data security, centralized access, and real-time updates.
- Overall, the solution provides a scalable, efficient, and user-friendly platform suitable for modern event management systems.

PROPOSED SYSTEM ARCHITECTURE

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Input:

Users enter login details, select events, provide booking information (name, email, number of tickets), and submit the booking through the web interface.

Processing:

The system verifies user input, performs validation, checks ticket availability, calculates total cost, and processes bookings using React for frontend and Node.js for backend logic.

Database:

All data such as users, event details, and booking records are stored and managed securely in the MySQL database.

Output:

The system displays event details, booking confirmation messages, updated ticket availability, and booking summaries dynamically on the web pages..

WORKFLOW



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❑ The user logs in and views available events.

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❑ The system retrieves event details and ticket availability from the database.

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❑ The user selects an event and enters booking details.

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❑ The system validates input and checks ticket availability.

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❑ The booking is processed and stored in the database.

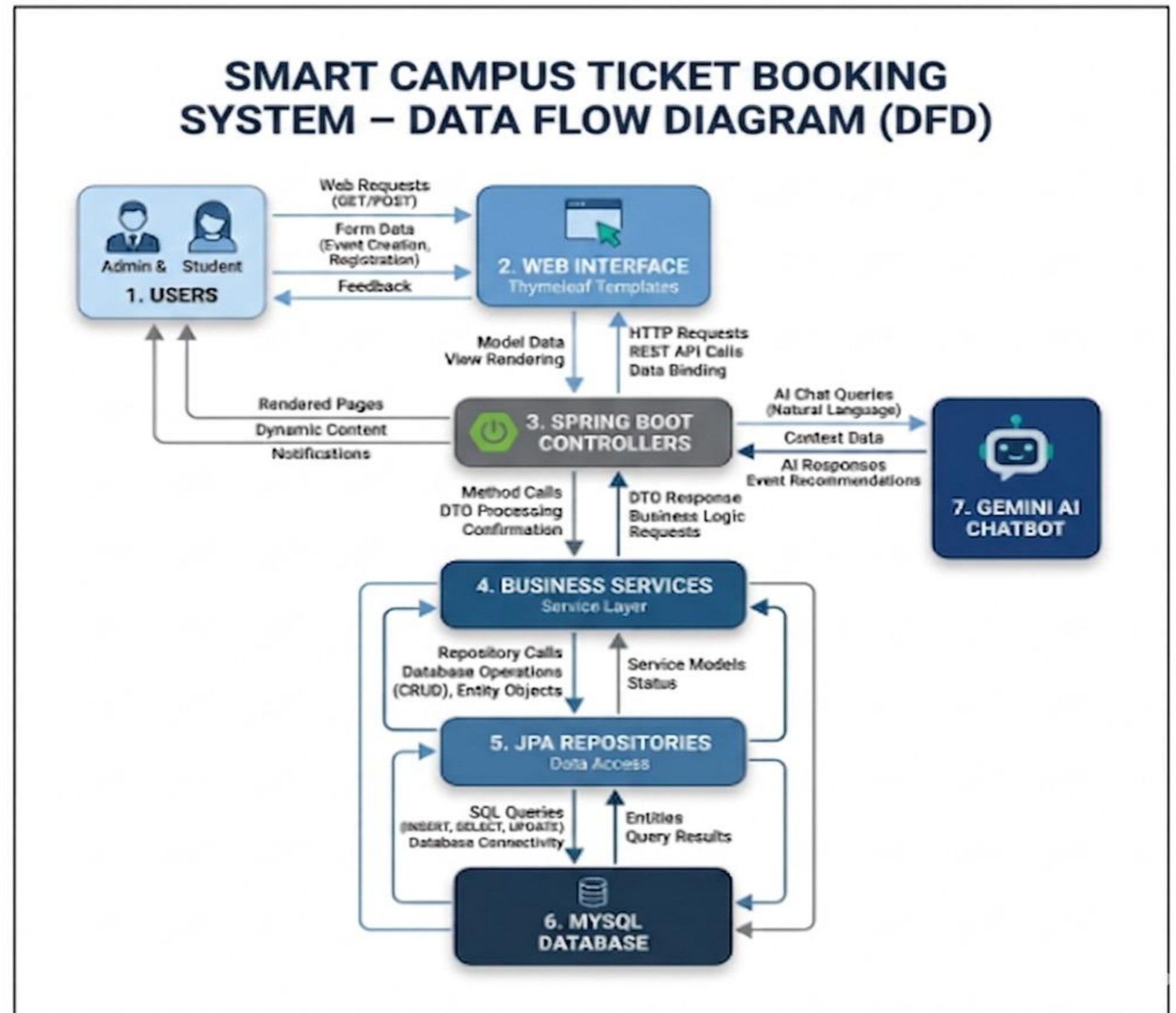
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❑ The admin can view and manage all bookings.

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❑ The system generates output such as booking confirmation and updated ticket count..

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TECHNOLOGIES USED

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Frontend:

 **React JS** – Used to build a dynamic and interactive user interface for login, event viewing, and ticket booking.

 **CSS3** – Used to design and style the website with a professional and responsive layout.

 **JavaScript** – Used for basic interactivity and improving user experience.

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Backend:

 **Node.js (Express)** – Used to handle server-side logic including authentication, booking processing, and API handling.

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Database:

 **MySQL** – Used to store and manage user details, product information, cart data, and order records securely.

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✂ Tools:

 **XAMPP** – Used as a local server environment integrating Apache and MySQL to run the project.

 **Visual Studio Code** – Used as the code editor for development.

 **Web Browser (Chrome/Edge)** – Used for testing and running the application.

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Software Requirements:

 Windows 10/11 Operating System

 XAMPP Server

 Web Browser

 Code Editor (Visual Studio Code)

- Node.js & npm

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MODULE DESCRIPTION



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Module 1: User Management

- Handles user login and basic authentication.
Validates user input such as email format.
- Manages user session using local storage.



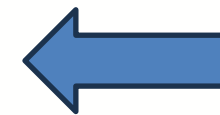
Module 2: Product Management

- Displays available events with details like, Name, date,time,venue, and price.
- Allows users to browse and select events easily



Module 3: Cart & Order Processing

- Stores user and booking data in MySQL database.
- Manages event details and ticket records.



Module 4: Database Management

- Handles ticket booking process .
- Collect user details.
- Calculates total cost and validates ticket availability.



IMPLEMENTATION SCREENSHOTS



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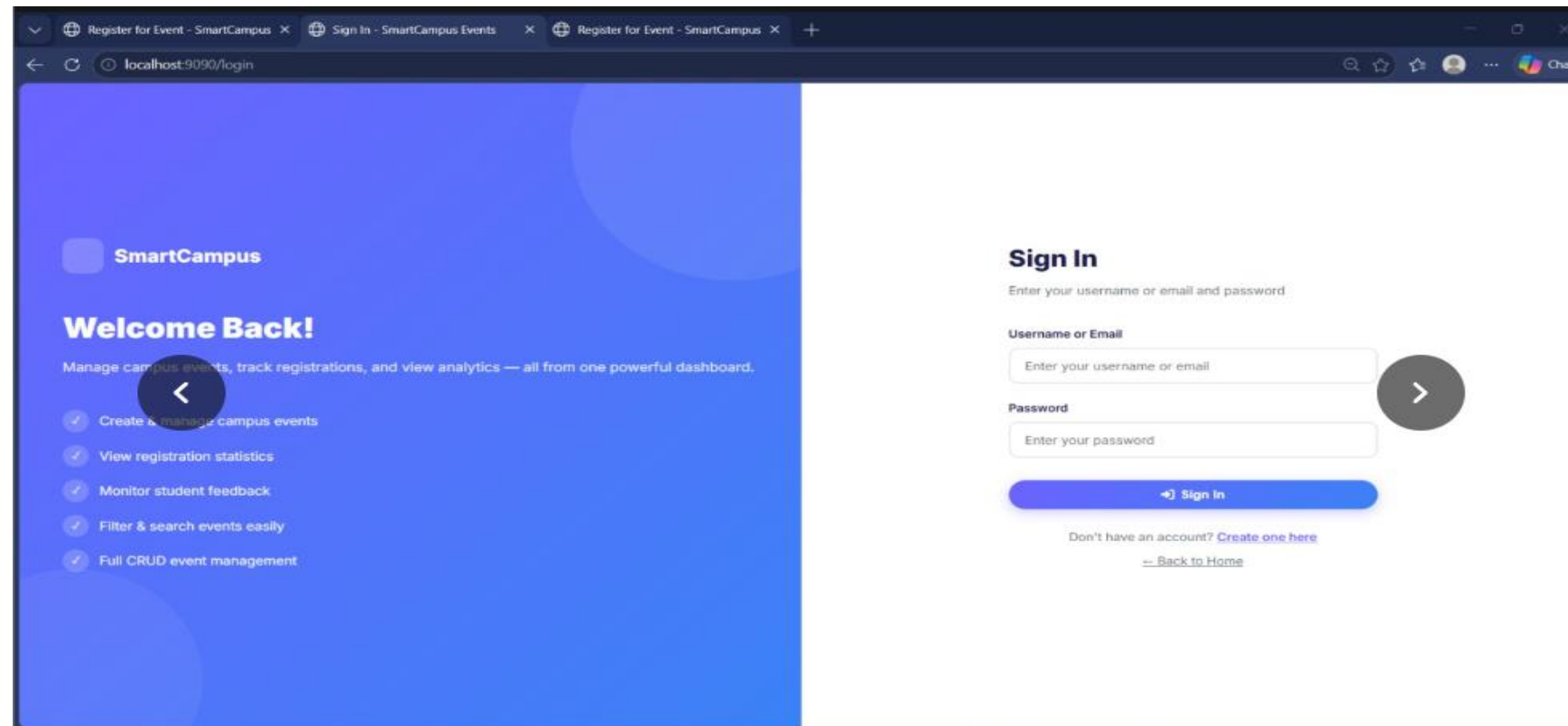
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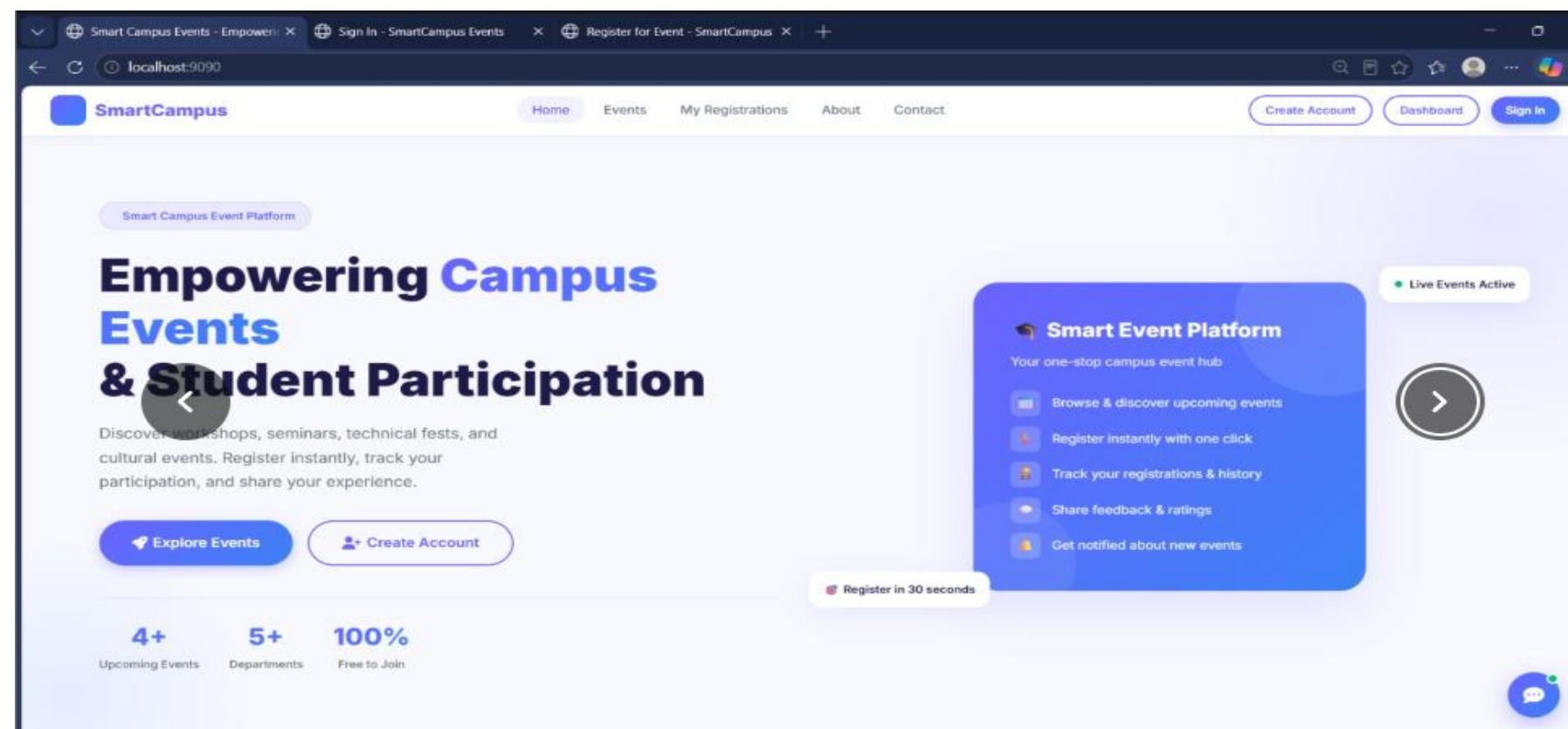
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Home Page – Event Dashboard

- Displays list of available events.
- Users can select events to view more details.
- The Admin dashboard option is available for authorized users.
- Provides simple and user – friendly navigation.



IMPLEMENTATION SCREENSHOTS

LOGIN & SIGNUP - User Authentication Module

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Create Account
Fill in your details to get started

Full Name
Enter your full name

Username
Min 4 characters, no spaces

Email Address
your@email.com

Account Type
Student

Password
Minimum 6 characters

Confirm Password
Re-enter your password

Create Account

Already have an account? [Sign in here](#)

- ✓ During login ,user enters email address.
- ✓ The ststem validates the email format.
- ✓ User details are checked using stored session /local storageing login, us.

- ✓ After successful login, the
- ✓ user is redirected to the Home page.
- ✓ Role-based is assigned



id	department	email	registered_at	student_name	tickets_booked	event_id
1	Computer Science	ushagopalakrishnan079@gmail.com	2026-04-20 15:47:41.000000	saraswathy	1	2
2	Computer Science	vu24364@veltech.edu.in	2026-04-21 00:40:40.000000	sandhya	1	5
3	Computer Science	ushagopalakrishnan079@gmail.com	2026-05-05 14:57:14.000000	saraswathy	1	1
4	Electronics	vu24364@veltech.edu.in	2026-05-05 15:27:53.000000	sandhya	1	3

IMPLEMENTATION SCREENSHOTS



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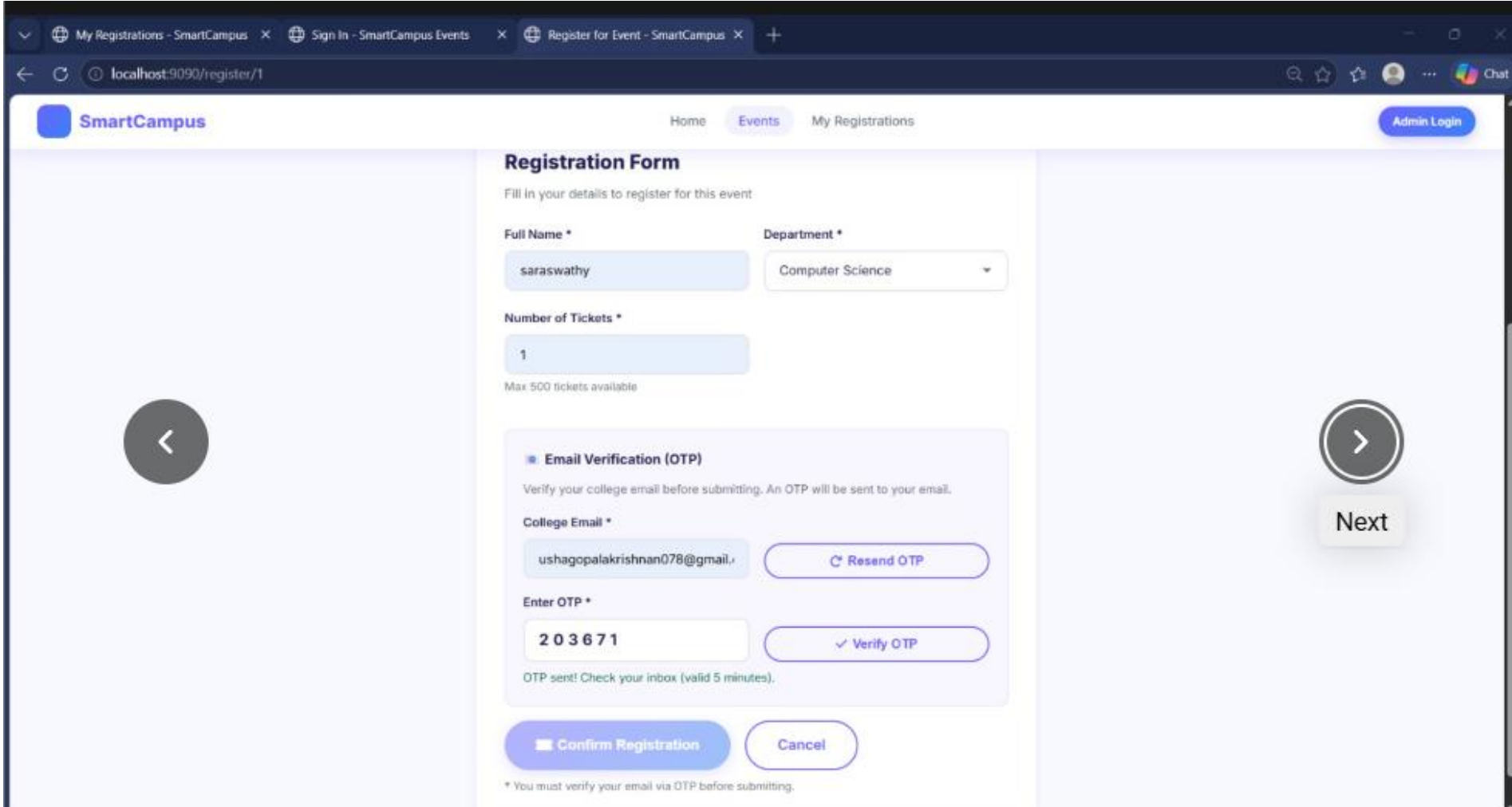
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8(B)

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SLIDE 3 – Event Booking Module

- ❖ When a user selects an event, the system displays event details including date, time, venue, and price.
- ❖ Users enter their name, email, and number of tickets for booking.
- ❖ The Event Booking Module allows users to select events and book tickets easily.
- ❖ The system validates input and checks ticket availability before confirming the booking.



The screenshot shows a web browser window with the URL `localhost:9090/register/1`. The page is titled "SmartCampus" and has navigation links for "Home", "Events", "My Registrations", and "Admin Login". The main content area is a "Registration Form" with the instruction "Fill in your details to register for this event". The form includes fields for "Full Name *" (containing "saraswathy"), "Department *" (a dropdown menu showing "Computer Science"), "Number of Tickets *" (a text input containing "1" with a note "Max 500 tickets available"), "College Email *" (containing "ushagopalakrishnan078@gmail.com" with a "Resend OTP" button), and "Enter OTP *" (containing "203671" with a "Verify OTP" button). Below these fields is a "Confirm Registration" button and a "Cancel" button. A message at the bottom states "OTP sent! Check your inbox (valid 5 minutes)." and a footer note says "* You must verify your email via OTP before submitting." The form is flanked by large circular arrows with a left arrow on the left and a right arrow on the right, with the word "Next" below the right arrow.

IMPLEMENTATION SCREENSHOTS



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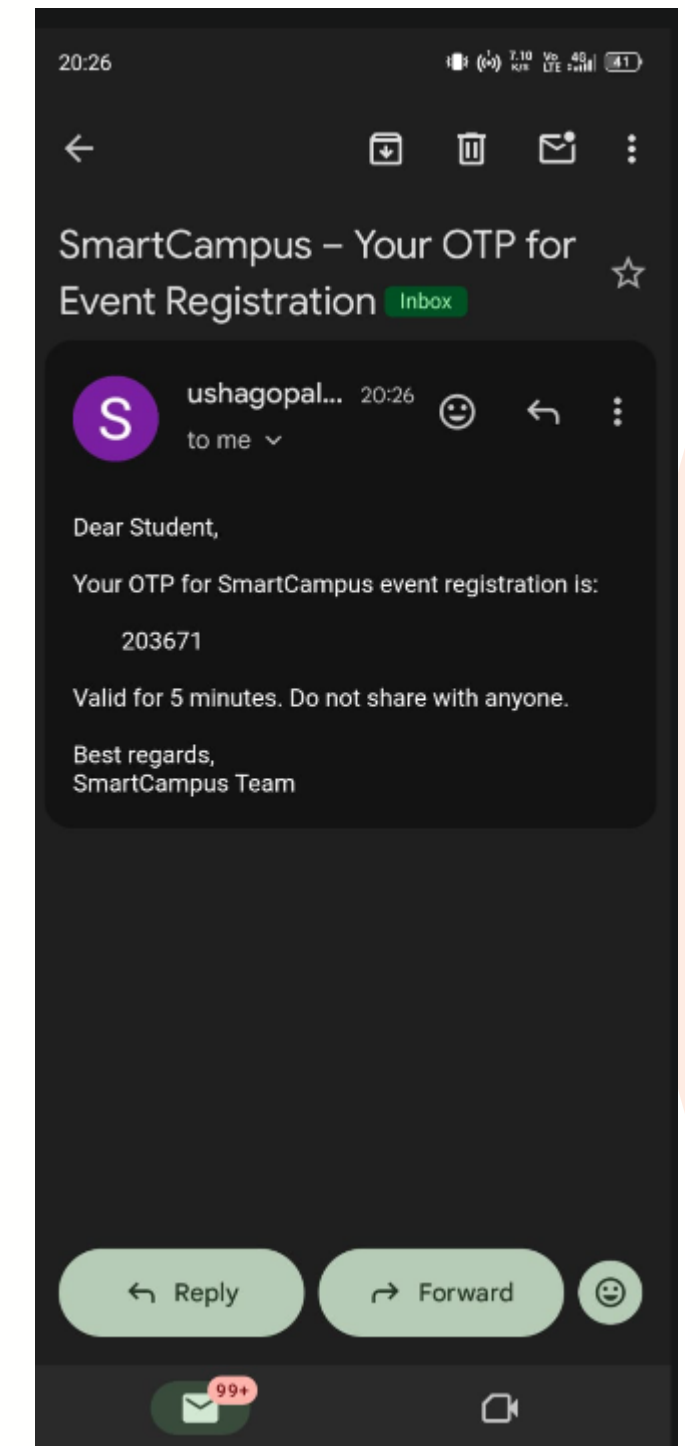
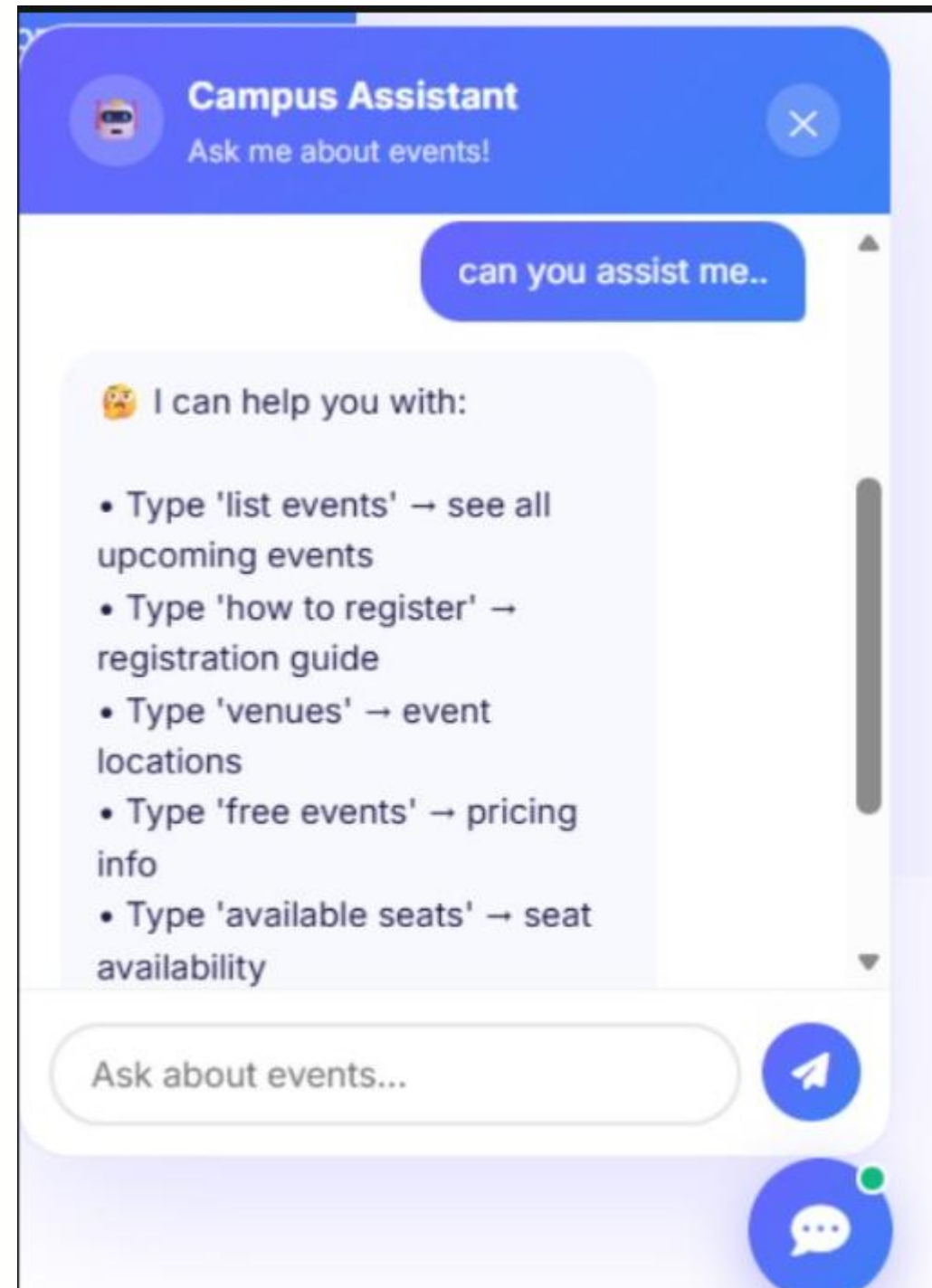
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8(B)

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SLIDE 3 – Event Booking Module

- ❖ AI-powered assistant to help users interact with the system conversationally
- ❖ Available 24/7 for instant support
- ❖ Reduces manual effort for admins and support teams
- ❖ The system validates input and checks ticket availability before confirming the booking.



IMPLEMENTATION SCREENSHOTS

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Admin Dashboard					
Total Bookings: 4					
← Back					
Download					
Name	Email	Event	Tickets	Total	Action
JANNATH	jannathyasmin73@gmail.com	Technical Fest 2026	5	₹2500	Delete
jk	jeonzara69@gmail.com	Technical Fest 2026	40	₹20000	Delete
tae	tae@gmail.com	AI & ML Seminar	4	₹1200	Delete
suga	suga12@gmail.com	AI & ML Seminar	1	₹300	Delete

The screenshot shows a web browser with multiple tabs. The active tab is 'localhost/phpmyadmin/index.php?route=/sql&pos=0&db=ticket_db&table=bookings'. The browser displays the phpMyAdmin interface for the 'ticket_db' database, showing the 'bookings' table. The table contains 4 rows of booking data. A 'Snipping Tool' window is open in the bottom right corner, displaying the text 'Screenshot copied to clipboard' and 'Automatically saved to screenshots folder.' The taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system clock in the bottom right corner shows '07:35' and '22-04-2026'.

id	name	email	event	tickets	total
1	JANNATH	jannathyasmin73@gmail.com	Technical Fest 2026	5	2500
2	jk	jeonzara69@gmail.com	Technical Fest 2026	40	20000
3	tae	tae@gmail.com	AI & ML Seminar	4	1200
4	suga	suga12@gmail.com	AI & ML Seminar	1	300

ORDERS & DATABASE - Order Output & Database Integration

- The Booking Page displays all booking details including user name, event name, tickets, and total amount.
- The booking data is stored in the database immediately after successful booking.
- This data is retrieved from the MySQL database using SQL queries.
- The backend (Node.js) connects and interacts with the database for storing and fetching booking records.

ADVANTAGES & LIMITATIONS



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Advantages

- ✓ Provides secure user login with input validation and role-based access.
- ✓ Reduces manual work by automating event booking and ticket management.
- ✓ EnsuresEnsures accurate data storage using MySQL database integration.
- ✓ Prevents overbooking through real-time ticket availability tracking.
- ✓ Enables centralized management of users, events, and bookings.

⚠ Limitations

- ❖ Currently designed for small-scale event management syatem (XAMPP).
- ❖ Does not include online payment gateway integration.
- ❖ Limited admin control features in the basic version.
- ❖ Requires internet connectivity for real-time deployment.

Thank you

